

Shikhar Srivastava

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Education

University of St Andrews, MSci. (Honours) Computer Science Sept 2023 – June 2028
• Student and Undergraduate Researcher for multiple continuous projects in Computer Science
• Skills Taught: Computer Architecture, Theory and Implementation of Algorithms, Object Oriented Programming, Systems-level Programming, Group Project Work, Functional Programming, Operating Systems Development, Programming Languages, Finite Machines

Experience

Project Intern, Tata Communications Transformation Services May 2024 – July 2024
• Migrated from YOLOv3 to transformer architecture for object detection using Hugging Face for Python
• Updated previous object detection solution to increase speed; increase accuracy from 65% to 75%, reduce training dataset and time.

Research in Constraint Programming and AI, University of St Andrews Sept 2024 – Present
• Added code to allow communication between compiler and SAT solvers in **Rust** and worked on proposals to implement divide-and-conquer concurrency in Abstract Symbol Tree simplifications.
• Product Owner in a running subproject that encodes all constraints into Conjunctive Normal Form to allow compilation from high-level constraint specification in **Essence**.

Photogrammetry from Archaeological Photographs, July 2025 – December 2025
University of St Andrews
• Worked with the School of CS at University of St Andrews as part of a Joint Project between the School of CS and the Griffith Institute at University of Oxford.
• Used Blender and Meshroom to create 3D models from archived photographs of Howard Carter's expedition to the Tomb of King Tut.
• Modelled King Tut's Death Mask for revamped expedition archive ([publicly released](#)).

Research Intern: VR and Mathematical Oncology, University of St Andrews February 2026 – May 2026
• Working with The Schools and Mathematics and Computer Science to create an interface for a Virtual Patient System, which will host Mathematical research on models which simulate cancerous tumour growth and metastases.

Deep Tech and AI Intern, Yotta Data Services June 2026 – August 2026
• Developed a Rust Library for Agent Harnesses, which offers between 3 and 5 times faster harnesses and tools over equivalent LangChain agents.
• Worked with Deep Tech and Agentic AI team to develop AI Agents and MCP servers that provide web UI quality assurance tools.

Projects

Conway's Game of Life [GitHub: Conway's GOL](#)
• Developed pure implementation of Conway's Game of Life in Haskell with variable sizes and parameters and keyboard-input; GUI with Gloss as Graphics Engine.

Hunt The Wumpus [GitHub: Hunt The Wumpus](#)
• Developed a keyboard-and-mouse based implementation of Hunt The Wumpus in Java, using Swing as a graphics engine. Features a GUI, customized ruleset and platform-independent implementation. Runnable using JDK.

Enigma Emulator [GitHub: Enigma](#)
• Built a digital twin of the Enigma Machine, directly analogous to the hardware. Provides encoding *and* decoding via the command line interface. Runnable on all ANSI-Compliant systems.

ferrOS: Research OS in functional rust (Ongoing) [GitHub: ferrOS](#)
• Building ferrOS, a research-oriented operating system, written in functional rust. Currently in a basic development stage, functional using nightly rust releases.

Skills

Languages and Technologies: Rust, C/C++, Java, Python, Haskell, PyTorch, Fusion 360, Blender

Soft Skills: Teamwork and group organisation, effective communication strategies, upskilling, SCRUM

Special Interests: Published author and photographic journalist (experience with student-run newspaper); Novelisation and Writing (self-publication); Wildlife Photography; Hiking; Birdwatching